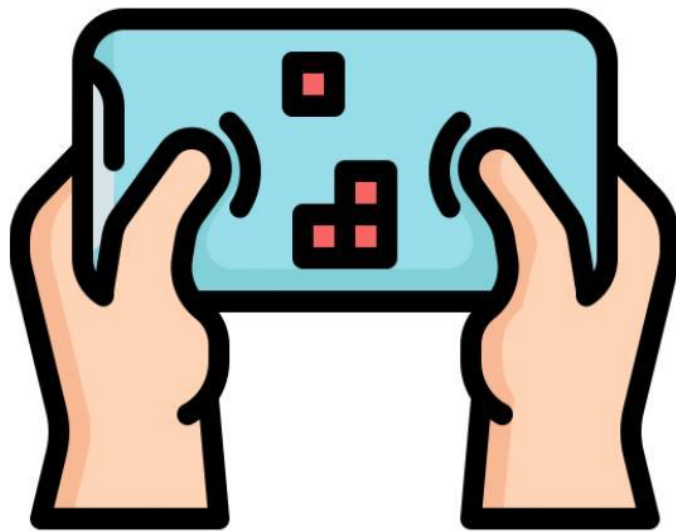


Game Design Document



1. Game Overview

1.1. Game Concept

A modernized, high-fidelity reimagining of the classic arcade Snake game set in a digital 'Grid-Sector'. Players control a bio-luminescent serpent that consumes energy nodes to grow. The game features a dynamic difficulty system where the simulation speed increases every 5 points, transforming a simple collection mechanic into a high-speed survival challenge.

1.2. Audience

Casual gamers, retro arcade enthusiasts, and players seeking reflex-based challenges. Suitable for all ages.

1.3. Platform / Technology

HTML5 (Canvas API), vanilla JavaScript, CSS3. Playable on modern desktop and mobile web browsers.

1.4. Genre

Arcade Action

1.4.1. Sub-Genre

Grid-based Puzzle / Survival

1.5. Camera / Perspective

2D Top-Down, fixed camera view centering on the 400x400 grid arena.

2. Gameplay

2.1. Objectives

Primary: Consume crimson energy nodes to achieve the highest possible score. Secondary: Survive increasing speeds without colliding with walls or the snake's own tail.

2.2. Progression

Linear progression based on score. The snake grows by one segment per node consumed. Game speed (tick rate) accelerates every 5 points consumed.

2.3. Play Flow

Start Game -> Navigate Grid -> Consume Food -> Grow & Accelerate -> Avoid Collision -> Game Over (display stats) -> Restart.

3. Mechanics

3.1. Rules

1. The snake moves continuously in the last input direction.
2. Consuming food adds +1 to score and length.
3. Colliding with grid boundaries (walls) ends the game.
4. Colliding with own body segments ends the game.
5. High score is persisted locally.

3.2. Physics

Discrete Grid Physics. Movement occurs on a 20x20 tile grid (400x400px total). Collision is checked frame-by-frame.

3.3. Character Movement

4-Directional controls (WASD or Arrow Keys). Features a 'zero-latency buffer' to allow precise turns. Spacebar toggles Pause state.

3.4. Economy

Score-based economy. No in-game currency. 'Food' acts as the pickup item.

4. Visual Graphics

4.1. Visual System

Cyberpunk/Neon aesthetic. Background is a linear gradient (Deep Violet #667eea to #764ba2). The snake is rendered in specific shades of Green (Emerald head, Forest-Green body) per user request and 'Shadow Glow' engine specs.

4.2. Interface (UI)

Minimalist HUD floating above the canvas. Semi-transparent glassmorphism effects on UI containers.

4.2.1. Game Menus

Main HUD: Current Score, High Score. Game Over Screen: Modal overlay with final score and 'Play Again' button.

5. First-Time User Flow / Experience

Immediate drop-in gameplay. Instructions ('Use Arrow Keys or WASD') are visible on-screen. No tutorial needed due to intuitive mechanics.

6. Story and Narrative

6.1. Backstory / Lore

The player enters the 'Grid-Sector', a digital vacuum where a bio-luminescent serpent thrives on energy nodes.

6.2. Main Plot

Survival in an infinite simulation loop.

6.2.1. Plot Progression

N/A - Arcade style loop.

6.3. Cutscenes

None.

7. Characters

7.1. Main Characters

The Neon Serpent: A green, glowing, segmented entity that elongates upon feeding.

7.2. Enemies

The Environment: The grid walls and the serpent's own growing tail act as the antagonists.

8. Game World

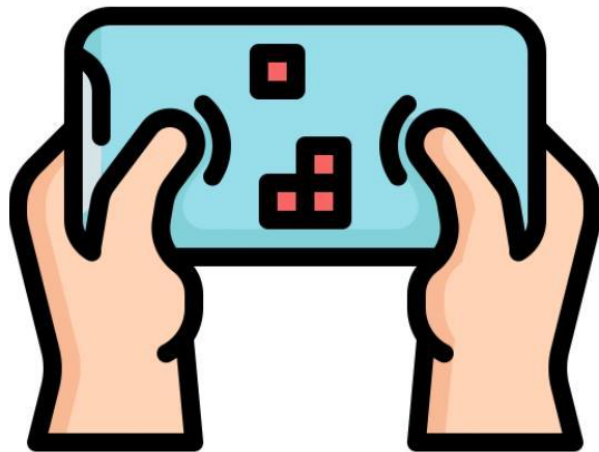
8.1. Look & Feel

Sleek, digital, and claustrophobic. Deep space colors provide contrast for the bright neon game elements.

8.2. Levels

Single procedural level: A 20x20 tile arena that effectively shrinks as the player occupies more space.

9. Art Assets



- Green Serpent segments (Procedural Rects)
- Crimson Energy Node (Food)
- Glassmorphism UI Containers
- Gradient Backgrounds
- System Fonts (Arial/Sans-serif)

10. Estimate Table

Scope	Hours	\$ Rate	Total Price
Art Work			
Game Development			
Testing			
Publishing			
Marketing (optional)			